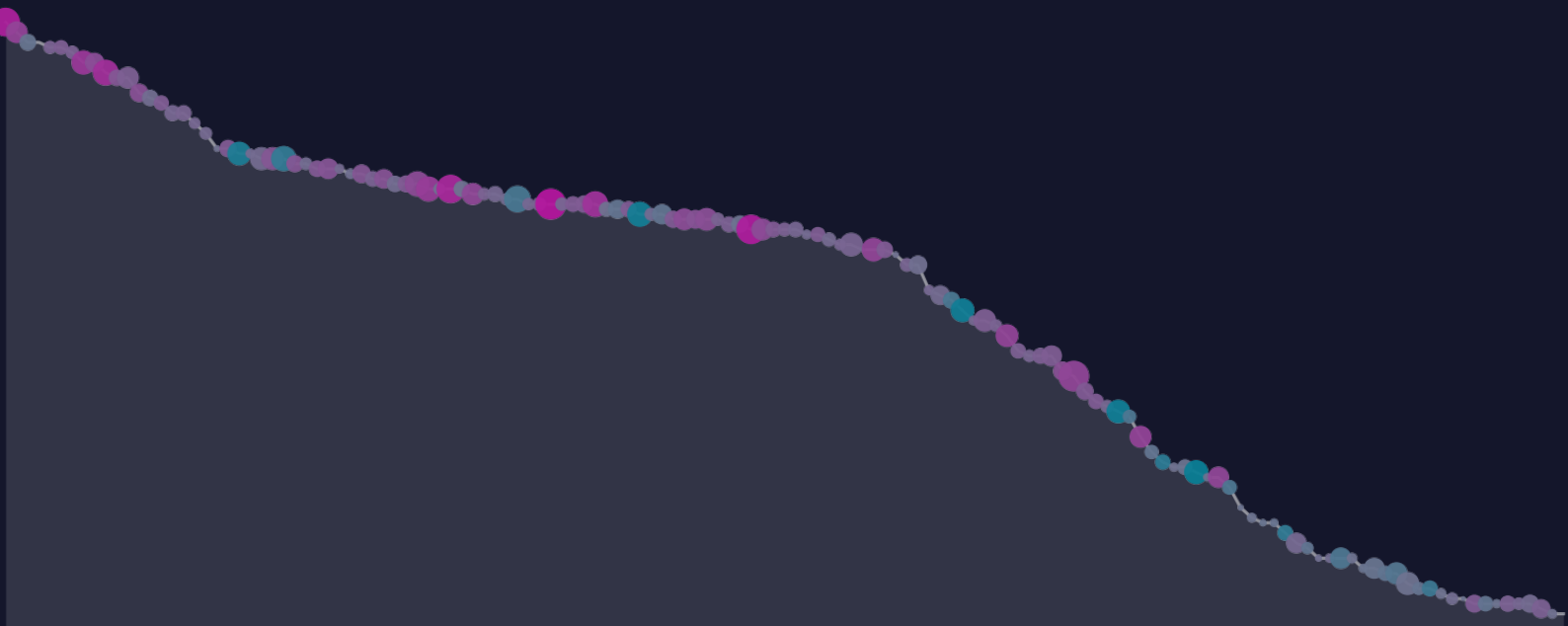


Where and when the GIRA network fails its users

Four years of station availability, reduced to a ranked list of which station needs a van, and at what hour.

Every station in the network, ordered by elevation. Colour is the direction of failure: magenta runs empty, cyan runs full. The slope is the finding.



AUTHORS

Rita Lima · Bruno Conceição
Cláudio Pitra · Erick Dos Santos

PREPARED FOR

EMEL Operations & Rebalancing

EVIDENCE

4,090,179 station snapshots

Contents

01	Introduction	3
	The system and the data	3
	The questions	3
	Summary of conclusions	4
02	Data and preparation	5
	Sources and the schema change	5
	Deduplication	5
	Flags and the analysis base	5
	Definitions used throughout	6
03	Analysis	7
	When does the network fail?	7
	Where does it fail, and does geography explain it?	9
	What should EMEL prioritise?	11
04	Conclusions and discussion	14
	Recommendations	14
	Limitations	14
	Further work	15
A	Cleaning decisions	16
B	Methodological notes	16
C	Supplementary tables	17
D	Reproducibility	19

The report in six numbers

5.24%

of snapshots have no bike available

6 hrs

later the empty peak falls at weekends

0.386

elevation vs failure direction, against
0.061 for latitude

33.3%

empty rate at the worst single station-
hour

1.11 km

median hop to the nearest station of
the opposite type

91 : 13

stations that run dry, against stations
that fill up

01

Introduction

GIRA does not fail on average. It fails at identifiable stations, at predictable hours, in a direction explained by the shape of the city.

1.1 The system and the data

SCOPE · Eight source files, January 2020 to March 2023. Availability snapshots only. No trip records exist in this data.

GIRA is the municipal bike-sharing system of Lisbon, operated by EMEL. It began in June 2017 as a pilot in Parque das Nações with 100 bikes and 10 stations, and by the first quarter of 2023 had grown to 147 active stations and roughly 2,900 docks, having passed ten million trips. Electric-assist bikes were part of the design from the outset, a decision driven by the city's topography. That topography turns out to be central to everything that follows.

The data underlying this report is *availability* data. Each row is a snapshot of one station at one moment, recording how many bikes were present and how many docks were free. Nothing in the source files records a journey, an origin or a destination. Every question that can legitimately be asked is therefore about availability: whether a rider arriving at a given station at a given hour would have found what they needed.

The audience for this work is EMEL's operations and rebalancing function. Their problem is not a shortage of bikes in the abstract. It is that they cannot tell *which* station, at *which* hour, should receive the next van. The deliverable that answers that is a ranked station-by-hour list, not a citywide average, and this report is organised to build towards it.

1.2 The questions

FAILURE, DEFINED · A station is *empty* when a rider who wants a bike finds none, and *full* when a rider who wants to return one finds no free dock. The two require opposite interventions.

Main question. Where and when does the GIRA network fail its users, and how should EMEL prioritise rebalancing and expansion?

- **SQ1. When?** How does the failure rate vary by hour, by day type, and by month across the four years?
- **SQ2. Where?** Which stations fail, in which direction, and does Lisbon's geography explain the pattern?
- **SQ3. What?** Which station-hours should be rebalanced first, can they be paired into short van loops, and did expansion happen where the failures are?

Both failure types are expressed throughout as rates, meaning the share of snapshots in a given bucket in which the condition held, and never as raw counts, because stations are logged at different frequencies and for different lengths of time.

1.3 Summary of conclusions

FOR THE SKIMMING READER · These six findings are the whole argument. Sections 3 and 4 supply the evidence for each; the appendices supply the workings.

1. **Failure is concentrated, not systemic.** The network is empty in 5.24% of snapshots and full in 2.04%. Those averages conceal station-hours above 30%.
2. **Failures follow commuters.** The empty rate peaks at 10:00 on weekdays but at 16:00 at weekends, a six-hour shift, and the full rate is its mirror image.
3. **Geography is the strongest single predictor.** Elevation correlates with the direction of failure at 0.386, against 0.061 for latitude.
4. **Many fixes are short.** The median distance from a worst empty-risk station to its nearest full-risk neighbour is 1.11 km.
5. **Expansion did not fix the worst stations.** The network nearly doubled, yet 479, 407, 452 and 309 remain problem stations.
6. **Supply, not distance, is the binding constraint.** Only 13 stations classify as full-risk against 91 empty-risk.

02

Data and preparation

Four years of files that do not share a schema, reduced to one comparable table, and one decision, about which rows belong in a denominator, that governs every rate in this report.

2.1 Sources and the schema change

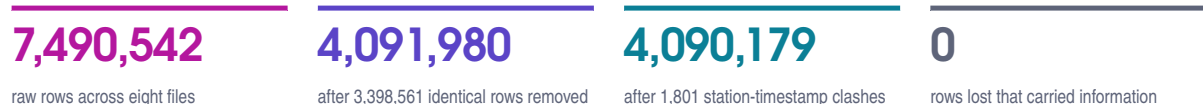
THE ROSETTA FILE · 2021 is the only year carrying both dock columns. The identity *bikes + empty docks = total docks* held on 100% of its 2.2 million rows, which is what licenses every harmonisation applied to the other years.

Eight files were supplied, covering 2020 by quarter, 2021 as a full year, 2022 by semester, and the first quarter of 2023. They total 7,490,542 rows and do not share a schema, and the differences are semantic rather than cosmetic. The 2020 files record `numdocasvacias`, the number of empty docks, and carry no station status column. The 2022 and 2023 files instead record `numdocas`, the total dock count, plus an `estado` status field.

Two further discrepancies matter. Timestamps in 2020 are Lisbon local time while 2021 onwards are UTC; both were converted to Europe/Lisbon, which leaves 146 rows falling in the daylight-saving clock-change gap with no valid local time. These become null by design, not by error, and are reported rather than silently dropped. Separately, `numdocas` means *working* docks in 2020 but *total physical* docks from 2021, so capacity comparisons across that boundary carry a caveat.

2.2 Deduplication

Cleaning removed 45.4% of the raw rows, which invites the obvious objection. The removal was almost entirely structural rather than corrective: the source system logs at fixed intervals, so when a station's state does not change between polls, a byte-identical row is written again. Removing those discards no information.



2.3 Flags and the analysis base

NOTHING OVERWRITTEN · Every correction was written to a new column or a boolean flag. The raw values remain available for validation, which is a course requirement and good practice regardless.

Three flags matter here. `numdocasvacias_fix` clips 116 negative dock counts to zero. `is_impossible` marks 20,976 rows (0.51%) where bikes exceed docks or capacity is zero. `is_repair`

marks 160,208 rows (3.92%) whose status indicates maintenance. The two overlap almost completely, since 20,864 of the impossible rows are also repair rows, so excluding both removes 160,320 rows rather than their arithmetic sum. Physically impossible readings are, in other words, mostly stations under maintenance.

The distinction that governs every number in this report is what those flags do to a rate. Both `is_empty` and `is_full` are already false for these rows, so they can never appear in a numerator. But they remain in the frame, and grouping over the whole dataset therefore leaves them in the *denominator*: a station under maintenance would be counted as working normally, and every rate would be diluted. All failure rates in Section 3 are computed on an analysis base that excludes them, comprising 3,929,859 rows or 96.08% of the cleaned data. The network-wide empty rate rises from 5.04% to 5.24% as a result, and the full rate from 1.96% to 2.04%.

A CAVEAT THAT MUST TRAVEL WITH THE TREND

Because 2020 has no `estado` column, no 2020 row can be flagged as under repair. The mask therefore removes 0.00% of 2020 but 6.31% of 2022 and 6.06% of 2023. Maintenance snapshots stay in the 2020 denominator and are removed from the later years, which makes 2020 look marginally better than the years it is compared against and very slightly steepens the upward trend in Section 3.1. The effect is bounded at 0.37 percentage points on rates near 5–6%. It is a limitation of the source data rather than a fixable defect, and it is measured rather than asserted away. *Quantified per year in Appendix B.1.*

Network growth is the one analysis that deliberately uses a different filter. A station under repair still physically exists and still has docks, so repair rows are retained when counting stations and capacity; only impossible rows are dropped.

2.4 Definitions used throughout

WHY A THRESHOLD AT ALL · Without one, a station observed a single time while empty scores 100% and tops every ranking. Six of 147 stations fall below it.

- **Empty rate.** Share of snapshots in a bucket with zero bikes available.
- **Full rate.** Share with zero free docks. **Failure rate** is the sum; the two conditions are mutually exclusive.
- **Tilt.** Empty rate minus full rate. Positive is empty-prone, negative full-prone. One signed number for a station's behaviour.
- **Failure type.** *Empty-risk* if at least 65% of a station's failures are empty events, *full-risk* if at most 35% are, *balanced* otherwise.
- **Reliable station.** At least 1,000 snapshots. Only these enter rankings.

03

Analysis

One sub-question at a time: when the network fails, where it fails, and what to do about it first.

Each subsection states the method used, the result, and the conclusion drawn, in that order. Context for the whole period is set by the network's growth: stations rose from 85 to 147 and docks from 1,568 to 2,928, with the sharpest jump between 2021 and 2022, when capacity grew faster than station count. Newer stations were built larger, not merely more numerous.

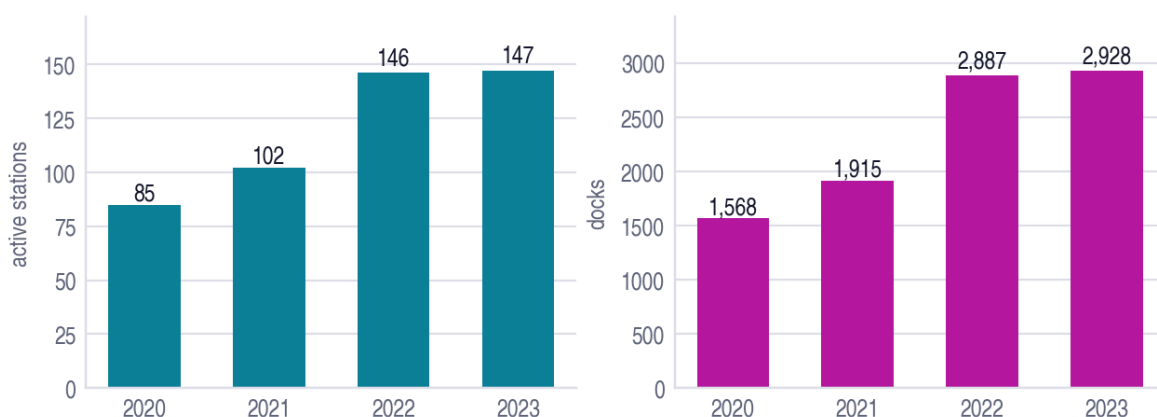


Figure 1. Network growth. Distinct active stations and total dock capacity by year, with capacity taken as one representative value per station rather than summed over rows. 2023 covers Q1 only and is not comparable to the full years beside it.

3.1 When does the network fail?

METHOD

RATES FROM BOOLEANS · Because `is_empty` is a boolean, its mean within a group is the rate for that group. No counting is involved.

Failure rates were computed on the analysis base by grouping on hour of day, then on the interaction of hour and day type, then on calendar month. The lockdown window drawn on Figure 4 is not hard-coded: it is derived from the minimum and maximum timestamps carried by the `period` column, so the shaded band cannot drift out of step with the cleaning.

ANALYSIS

The two failure directions are near-perfect mirror images across the day. The empty rate bottoms out at 3.82% around 07:00, climbs sharply through the morning, and holds a plateau between 6.2% and 6.8% from 10:00 to 18:00, peaking at 6.83% at 11:00. The full rate does the opposite, reaching 3.74% before dawn and falling below 1% by early afternoon.

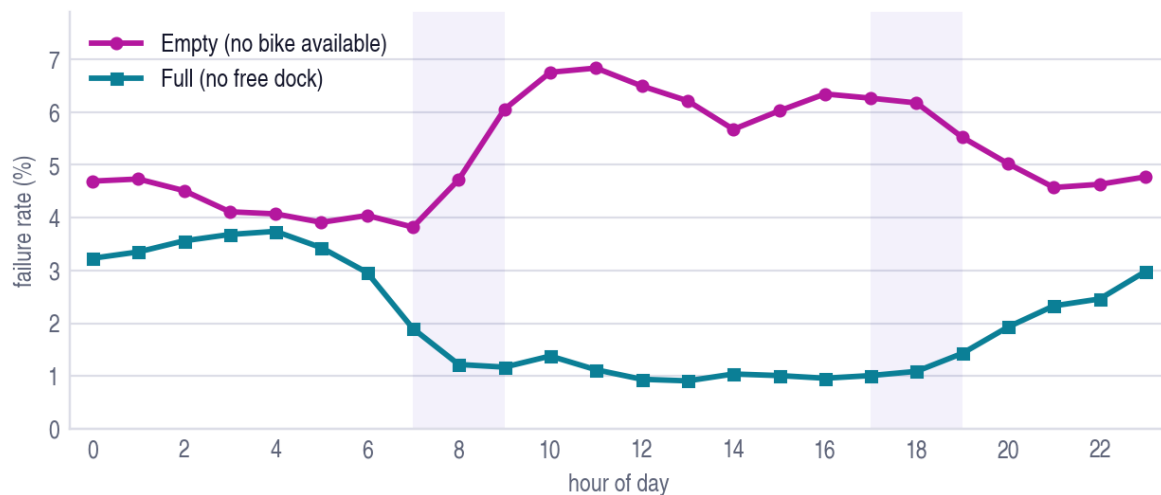


Figure 2. The daily cycle. Failure rate by hour, analysis base; shaded bands mark the commute windows. Bikes drain out of some stations during the day and pile back up overnight, which is the signature of a distribution problem rather than a fleet-size one.

Splitting by day type turns that pattern from a description into a mechanism. If failures were random with respect to human schedules, weekdays and weekends would look alike. They do not. The weekday empty rate peaks at 10:00 at 7.75%; the weekend empty rate peaks at 16:00 at 7.02%, a full six hours later. Weekdays also swing harder, from 3.9% to 7.7% against 3.4% to 7.0% at weekends. A sharp morning weekday spike and a later, flatter weekend curve is what commuting and leisure demand look like when separated.



Figure 3. The commuter fingerprint. Empty and full rates by hour, split by day type. The six-hour displacement of the weekend empty peak is the clearest single piece of evidence that the network fails on a schedule rather than at random.

READ WITH CARE · Availability improved in April 2020 because demand collapsed, not because service improved. The failure rate is a demand signal as much as a service signal.

The monthly series adds a caution about interpreting the rate at all. April 2020 records the lowest failure rate in the entire period, 1.34%, during the Portuguese state of emergency. Availability was not good that month because the service was working well; it was good because almost nobody was riding, so every bike stayed docked. Recovery took roughly four months, and by September 2020 the rate had reached 8.44%, comfortably above the 3.82% and 4.49% of January and February. March 2020 appears misleadingly high at 7.66% because the emergency was declared on the 18th, so the month mixes two very different halves.

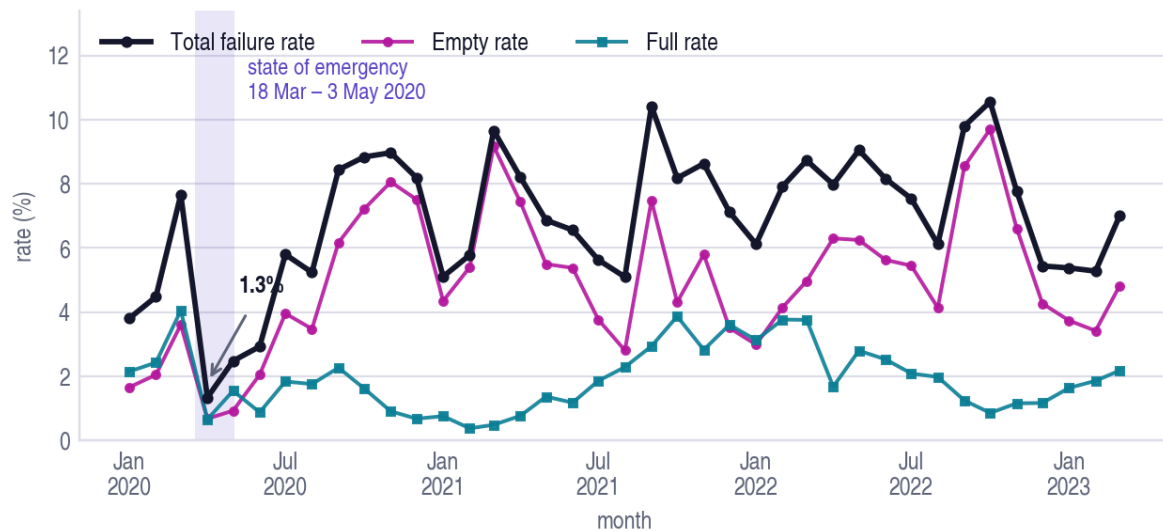


Figure 4. Four years, one demand signal. Monthly failure rate with the state of emergency shaded. The peak of the whole series is October 2022 at 10.56%. The rise through 2021 and 2022 partly reflects a busier, better-used network rather than a degrading one.

CONCLUSIONS

The network fails on a predictable, schedulable rhythm driven by commuting: empty by day and full by night on weekdays, shifted six hours later at weekends. Because the pattern is predictable, it is addressable by scheduling rather than by capital expenditure. The monthly series also establishes that the failure rate should never be read as a service-quality score on its own.

3.2 Where does it fail, and does geography explain it?

METHOD

Each station was profiled on the analysis base, yielding its empty and full rates, median coordinates, a representative capacity and its tilt. Of 147 stations profiled, 141 met the 1,000-snapshot threshold and only those enter the rankings. The hills hypothesis was tested twice: first with latitude as a cheap proxy for "further north is further uphill", and then with true elevation joined per station from the Open-Meteo elevation API. Testing the proxy first matters, because it establishes what the second test adds.

ANALYSIS

EFFECT SIZE · Fitting a line, every additional 10 m of elevation is worth roughly 0.45 percentage points of empty rate.

The proxy fails. Tilt correlates with latitude at only 0.061, and the median latitudes of empty-prone and full-prone stations differ in the fourth decimal place. Real elevation is a different story: tilt correlates with elevation at 0.386, and the raw components move in opposite directions, with empty rate at 0.365 and full rate at -0.176 . Empty-prone stations sit at a median elevation of 81 m against 29.5 m for full-prone ones.

WHY THIS IS THE STRONGEST FINDING

The direction of failure is not distributed at random across the city. It is organised by gravity. Riders coast downhill towards the river and do not ride back up, so hillside stations drain and riverside stations overflow. This also explains why buying more bikes would not fix it: the bikes are not missing, they are in the wrong place, and they got there for a reason that will repeat tomorrow.



Figure 5. The failure map. Every reliable station by coordinate. Colour is tilt, size is total failure rate. The magenta cluster sits on high ground north and west of the centre; the cyan sits on the riverfront and in the eastern Parque das Nações corridor.

Under the three-way classification the imbalance is stark: 91 stations are empty-risk, 37 balanced, and only 13 full-risk. The individual extremes are correspondingly lopsided.

TABLE 1 The five worst stations in each failure direction, reliable stations only.

EMPTY-RISK STATION	EMPTY %	FULL-RISK STATION	FULL %
479 - Rua Professor Oliveira Marques	21.89	452 - Rua Teixeira de Pascoais	10.79
407 - Praça Duque de Saldanha	19.90	309 - Rua da Palma	10.60
364 - Rua D. Francisco M. de Melo	19.17	316 - Rua António Pedro	10.15
412 - Av. Ressano Garcia	18.11	311 - Avenida Almirante Reis 14	10.11
314 - Rua Marques da Silva	16.25	552 - Alameda das Linhas de Torres	9.71

PERSISTENCE, PRECISELY - Spearman $\rho = 0.388$ between matched first quarters, with 4 of the worst 10 stations still in the worst 10 three years later. The claim is not that nothing improved, only that the worst offenders were not fixed.

Two secondary checks qualify the picture. Major transit interchanges (Oriente, Entrecampos, Cais do Sodré, Saldanha and Santa Apolónia) do not form a distinct failure class: their mean failure rate of 7.43% is barely above the network's 7.17%, though they are larger, averaging 25.4 docks against 19.6. And comparing matched first quarters, 2020 against 2023, the mean failure rate across the 81 comparable stations actually fell slightly, from 5.56% to 5.05%.

CONCLUSIONS

Lisbon's topography, not its street layout or its north-south axis, writes the failure map. Elevation should be treated as a planning variable: it predicts which direction a station will fail in before a single snapshot is collected, which makes it useful for siting as well as for rebalancing. It is one driver among several, since transit access and land use plainly matter too, which is why the correlation is 0.386 rather than 0.9, but it is the strongest one available here.

3.3 What should EMEL prioritise?

METHOD

The unit of analysis becomes the station-hour, because that is what a van is dispatched to. Rates were computed for every station-hour cell with at least 30 observations, giving 3,384 usable cells. A raw failure rate alone is a poor priority score, because a station empty at 04:00 harms almost nobody while the same rate at 18:00 harms many. Cells were therefore weighted by a commute-shaped demand curve.

AN ASSUMPTION THAT MUST BE OWNED

The demand curve is *not* derived from this dataset. Availability snapshots record how full a station is, not how many people wanted a bike, and deriving demand from the same data would be circular with the failure rate itself. A fixed curve was used instead, shaped to the commuting pattern established in Section 3.1. The ordering at the top of the priority list is stable for any curve peaking at the commute hours, and the unweighted ranking stands on its own if the assumption is rejected. *Curve tabulated in Appendix B.3.*

ANALYSIS

The demand-weighted ranking is unambiguous at the top. Station 519, Rua Prof. Pulido Valente, at 112 m, is empty in 33.3% of its 09:00 snapshots, the single highest-priority intervention in the network. It is followed by 364 and 407 in the evening peak, and by station 314, which appears repeatedly across the middle of the day and reaches 40.5% empty at 12:00.

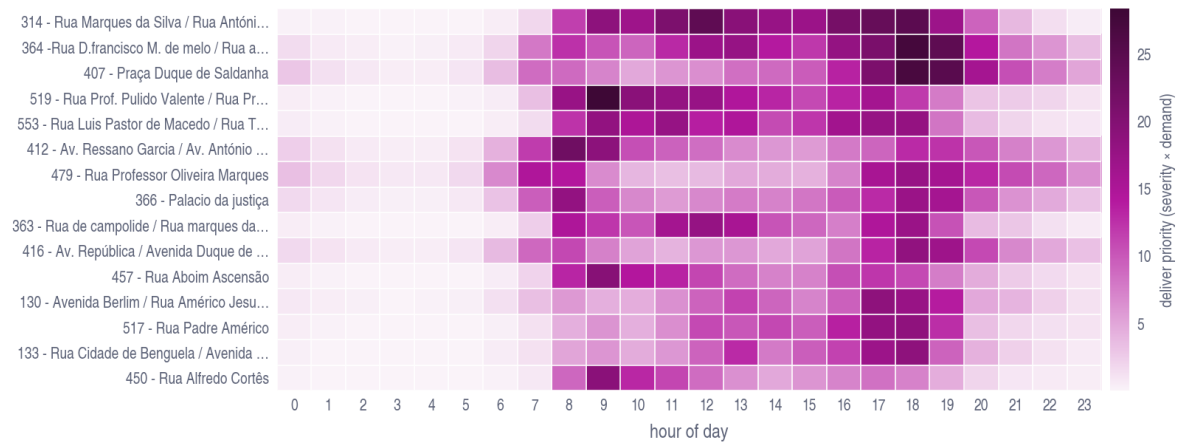


Figure 6. Where to deliver. The fifteen highest-priority stations for *delivering* bikes, by hour, coloured by severity weighted by demand. The bright band from mid-morning to the evening peak is the daily window in which the network is short of bikes uphill.

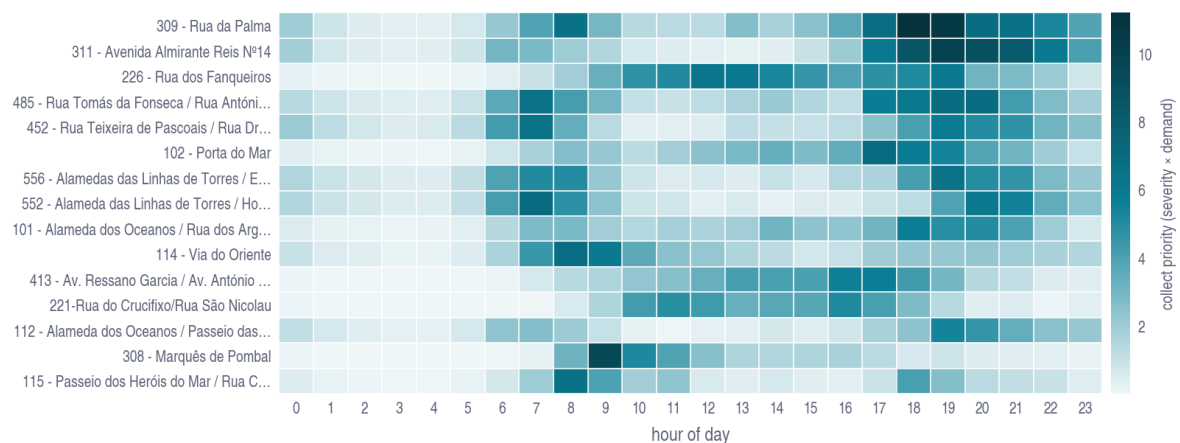


Figure 7. Where to collect. The fifteen highest-priority stations for *collecting* bikes. Station 309, Rua da Palma, at 31 m, tops the list and peaks in the evening, being the accumulation point for bikes ridden downhill during the day.

WHY GREAT-CIRCLE DISTANCE · A degree of longitude is shorter than a degree of latitude, and more so away from the equator. Raw coordinate differences would systematically misjudge east–west distance at Lisbon's latitude.

Pairing the two directions inside a shared time window produces concrete van routes. Each source station is used once, and its partner is chosen by true great-circle distance.

TABLE 2 Suggested van loops for the two peak windows. Positive climb means the van performs the ascent that riders decline to make.

WINDOW	COLLECT FROM	DELIVER TO	KM	CLIMB
Morning	552 - Alameda das Linhas de Torres	519 - Rua Prof. Pulido Valente	1.07	+27 m
Morning	308 - Marquês de Pombal	412 - Av. Ressano Garcia	1.18	+33 m
Morning	114 - Via do Oriente	457 - Rua Aboim Ascensão	5.95	+72 m
Evening	309 - Rua da Palma	407 - Praça Duque de Saldanha	1.86	+48 m
Evening	311 - Avenida Almirante Reis	364 - Rua D. Francisco M. de Melo	2.21	+77 m
Evening	485 - Rua Tomás da Fonseca	314 - Rua Marques da Silva	4.18	-44 m

Whether this is a rebalancing problem or a sizing problem was tested directly, by measuring how far each of the fifteen worst empty-risk stations sits from its nearest full-risk neighbour. Distance is not the obstacle: the median hop is 1.11 km and 7 of the 15 are within a kilometre. Supply is. With only 13 full-risk stations serving 91 empty-risk ones, a single station, 316 Rua António Pedro, is the nearest full-risk partner for ten of the fifteen. One station cannot supply ten.

Finally, expansion was checked against failure. Of the 141 reliable stations, 85 date from 2020, 18 first appear in 2021 and 38 in 2022. The 2021 cohort fails *more* often than the original network, at 8.16% against 7.04%, and the 2022 cohort matches it at 7.02%. Growth added capacity in places that fail at least as much as the places that already existed.

CONCLUSIONS

EMEL should treat the demand-weighted station-hour ranking as a standing daily to-do list, beginning with station 519 at 09:00 and the Saldanha cluster in the evening peak. The short hops mean a modest number of van loops would address a disproportionate share of the failures. But the rebalancing answer has a ceiling, and the network has already reached the edge of it: with a seven-to-one imbalance between stations that run dry and stations that fill up, there is not enough naturally accumulating supply to redistribute, and the remainder is a question of dock capacity and station siting.

04

Conclusions and discussion

Three answers, five recommendations, and four limitations that a reader is entitled to know before acting on any of it.

4.1 Answers to the questions

The answer to *when* is that failure follows the commute: empty by day and full overnight on weekdays, displaced six hours later at weekends, with a monthly series that tracks demand rather than service quality. The answer to *where* is that failure is organised by altitude, with uphill stations running dry and riverside stations overflowing, at a correlation of 0.386 against 0.061 for latitude. The answer to *how to prioritise* is the demand-weighted station-hour ranking in Section 3.3, which converts those two findings into a dispatch order.

The through-line is that the network does not fail on average. It fails at identifiable stations, at predictable hours, in a direction explained by gravity. Each of those three properties makes the problem more tractable than a citywide average would suggest.

4.2 Recommendations

1. **Adopt the station-hour ranking as the daily dispatch list**, rather than a station-level ranking. The distinction is not cosmetic: station 314 is unremarkable overall but exceeds 40% empty at midday.
2. **Run the two peak windows as fixed loops**. The first two morning pairs in Table 2 are under 1.2 km, so these are short, repeatable movements rather than improvisation.
3. **Use elevation as a siting criterion**. It predicts failure direction before any operating data exists, which makes it usable at the planning stage rather than in hindsight.
4. **Treat the full-risk shortage as a capacity question**. Thirteen accumulating stations cannot supply ninety-one draining ones. Additional docks at riverside and downtown accumulation points would increase the supply available to rebalance from.
5. **Re-examine the 2021 expansion cohort**. It fails more than the network it was added to, which suggests siting decisions in that round deserve review.

4.3 Limitations

STATED, NOT BURIED · Each limitation is quantified where it can be, and each is signposted from the section it affects.

- **No trip data**. Everything here is inferred from availability snapshots. Actual rider demand, and therefore unmet demand, is not observable in this dataset.

- **The demand curve is assumed.** The priority ranking depends on a fixed commute-shaped weighting that is not derived from the data.
- **The repair flag does not exist for 2020**, so the analysis mask is applied unevenly across years and mildly flatters 2020, bounded at 0.37 percentage points.
- **2023 covers one quarter**, so year-on-year comparisons involving it are made on matched first quarters rather than full years.

A fifth point is interpretation rather than data quality: the failure rate rises across the period, but as the lockdown episode demonstrates, that metric moves with ridership. Part of the rise through 2021 and 2022 reflects a busier network, not a deteriorating one.

4.4 Further work

Three extensions would sharpen this materially. Trip-level data would replace the assumed demand curve with a measured one and allow unmet demand to be estimated rather than inferred. Weather data would test a plausible confound, since rain suppresses cycling unevenly across a hilly city and may interact with the elevation finding. And a real drive-time matrix, rather than great-circle distance, would turn the suggested loops in Table 2 into an operational routing schedule that accounts for traffic and one-way streets.

A-D

Appendices

The workings: what was cleaned and why, which thresholds were chosen and on what grounds, the supporting tables, and how to reproduce all of it.

Appendix A · Cleaning decisions

Every decision below was applied in the cleaning notebook, and in every case the original value was preserved and the correction written to a new column.

TABLE A.1 Cleaning log.

ISSUE	TREATMENT	ROWS
Byte-identical duplicate rows	Dropped	3,398,561
Duplicate station + timestamp	First occurrence kept	1,801
Negative empty-dock counts	Clipped to 0 in <code>numdocasvacias_fix</code>	116
Bikes > docks, or zero capacity	Flagged <code>is_impossible</code>	20,976
Station under maintenance	Flagged <code>is_repair</code>	160,208
DST clock-change gap	Local timestamp null by design	146
Corrupted 2020 station label	Repaired in <code>desigcomercial_clean</code>	n/a
Station identity across years	Numeric prefix to <code>station_id</code>	all

AN ORDERING TRAP · Computed before flagging, zero-capacity stations divide by zero and produce infinities that propagate into every downstream mean without raising an error.

One ordering decision is worth recording because it silently corrupts averages: the occupancy rate must be computed *after* impossible rows are flagged, and set to null for those rows immediately afterwards.

Appendix B · Methodological notes

B.1 IMPACT OF THE ANALYSIS MASK, BY YEAR

This quantifies the caveat raised in Section 2.3. The shift column is the point: small in every year, but not evenly distributed, because the repair flag does not exist for 2020.

TABLE B.1 Rows and rates before and after applying the analysis mask.

YEAR	RAW ROWS	BASE ROWS	% DROPPED	EMPTY BEFORE	EMPTY AFTER	SHIFT
2020	398,163	398,153	0.00	3.86	3.86	0.00
2021	1,232,857	1,226,155	0.54	5.40	5.43	+0.03
2022	1,852,486	1,735,608	6.31	5.47	5.84	+0.37
2023	606,527	569,797	6.06	3.74	3.98	+0.24

As a consistency check, the excluded rows show an empty rate and a full rate of exactly 0.000%, which is what the flag definitions require. Their median bike count is zero, consistent with stations out of service rather than merely busy.

B.2 THRESHOLDS AND WHY THEY WERE CHOSEN

- **1,000 snapshots for station reliability.** Far below the median of 35,535 while excluding the genuinely thin cases. Six stations are excluded; the least observed survivor has 3,463 snapshots.
- **30 observations per station-hour cell.** A lower bar, because a single hour holds proportionally fewer snapshots. 3,384 cells qualify.
- **65% / 35% for the three-way failure type.** A simple majority rule would label 107 stations empty-prone and 34 full-prone; the stricter rule gives 91, 37 and 13.
- **Weighted station averages.** Station failure rates are weighted by snapshot count, since an unweighted mean across hours would treat 03:00 as equal to 18:00.

B.3 THE DEMAND CURVE

Relative rider activity by hour, on a 0–1 scale, used to weight severity in Section 3.3. This is an assumption, not a measurement.

TABLE B.2 Assumed relative demand by hour.

HOUR	0	1	2	3	4	5	6	7	8	9	10	11
Weight	.10	.05	.03	.02	.02	.05	.20	.55	.95	.85	.55	.55
HOUR	12	13	14	15	16	17	18	19	20	21	22	23
Weight	.60	.60	.55	.55	.70	.95	1.00	.85	.55	.40	.28	.18

Appendix C · Supplementary tables

C.1 FAILURE RATE BY HOUR OF DAY

TABLE C.1 Empty and full rates (%) by hour, analysis base.

HR	EMPTY	FULL	HR	EMPTY	FULL	HR	EMPTY	FULL	HR	EMPTY	FULL
0	4.69	3.23	6	4.04	2.96	12	6.49	0.94	18	6.17	1.09
1	4.73	3.35	7	3.82	1.90	13	6.21	0.91	19	5.52	1.43
2	4.51	3.56	8	4.72	1.22	14	5.67	1.04	20	5.02	1.93
3	4.11	3.68	9	6.05	1.17	15	6.02	1.01	21	4.57	2.33
4	4.07	3.74	10	6.75	1.38	16	6.34	0.96	22	4.63	2.46
5	3.91	3.43	11	6.83	1.12	17	6.26	1.01	23	4.77	2.98

C.2 MONTHLY FAILURE RATE

TABLE C.2 Monthly empty, full and total failure rates (%).

MONTH	EMPTY	FULL	TOTAL	MONTH	EMPTY	FULL	TOTAL
2020-01	1.66	2.15	3.82	2021-09	7.48	2.94	10.43
2020-02	2.05	2.43	4.49	2021-10	4.31	3.87	8.18
2020-03	3.61	4.05	7.66	2021-11	5.80	2.83	8.63
2020-04	0.68	0.66	1.34	2021-12	3.51	3.61	7.12
2020-05	0.93	1.55	2.48	2022-01	2.99	3.13	6.12
2020-06	2.05	0.88	2.93	2022-02	4.15	3.77	7.92
2020-07	3.96	1.84	5.79	2022-03	4.97	3.76	8.73
2020-08	3.48	1.76	5.24	2022-04	6.30	1.68	7.98
2020-09	6.17	2.27	8.44	2022-05	6.25	2.80	9.05
2020-10	7.22	1.62	8.84	2022-06	5.63	2.53	8.16
2020-11	8.06	0.91	8.97	2022-07	5.45	2.09	7.54
2020-12	7.51	0.68	8.19	2022-08	4.14	1.98	6.12
2021-01	4.35	0.76	5.11	2022-09	8.55	1.25	9.80
2021-02	5.40	0.38	5.79	2022-10	9.70	0.86	10.56
2021-03	9.17	0.49	9.65	2022-11	6.61	1.16	7.77
2021-04	7.44	0.77	8.20	2022-12	4.27	1.17	5.44
2021-05	5.50	1.37	6.87	2023-01	3.74	1.64	5.38
2021-06	5.38	1.18	6.56	2023-02	3.42	1.86	5.28
2021-07	3.77	1.86	5.63	2023-03	4.82	2.18	7.00
2021-08	2.81	2.29	5.10				

C.3 TOP DEMAND-WEIGHTED PRIORITIES

TABLE C.3 The ten highest-priority station-hours for delivering bikes.

STATION	HOUR	EMPTY %	ELEV.	PRIORITY
519 - Rua Prof. Pulido Valente	09	33.3	112 m	28.3
364 - Rua D. Francisco M. de Melo	18	27.4	120 m	27.4
407 - Praça Duque de Saldanha	18	27.0	79 m	27.0
407 - Praça Duque de Saldanha	19	29.4	79 m	24.9
314 - Rua Marques da Silva	18	24.7	50 m	24.7
364 - Rua D. Francisco M. de Melo	19	28.7	120 m	24.4
314 - Rua Marques da Silva	12	40.5	50 m	24.3
314 - Rua Marques da Silva	17	24.6	50 m	23.4
412 - Av. Ressano Garcia	08	23.5	87 m	22.3
314 - Rua Marques da Silva	16	30.7	50 m	21.5

Appendix D · Reproducibility

TWO-STAGE DESIGN · A cleaning notebook writes `gira_clean.parquet` ; the analysis notebook reads only that file and never modifies it.

All results were produced in Python 3.13 using pandas, matplotlib, seaborn, folium and scipy, from a cleaned dataset stored as Parquet. The analysis notebook runs end to end from a fresh kernel; execution counts in the submitted copy run 1 to 32 without gaps, and every figure and table in this report traces to a numbered cell in it. Elevation is the single external input, fetched once per station from the Open-Meteo elevation API in batches of one hundred coordinates.

The analysis base, which determines the denominator of every rate in Section 3, is defined in a single place and reused throughout:

```
base = gira[~gira['is_repair'] & ~gira['is_impossible']].copy()
```

An interactive Folium map of the failure geography, the static counterpart of which is Figure 5, is written to `gira_failure_map.html` and opens in any browser without a server.